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Course: CS683

Assignment: 1

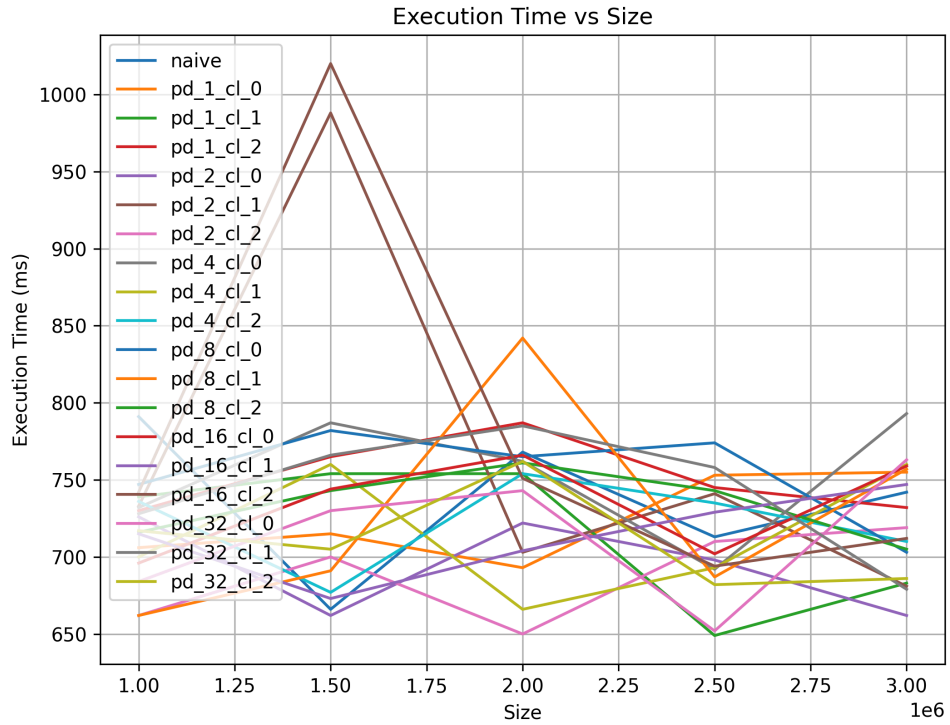
Task: 2

Task 2 (8 points)

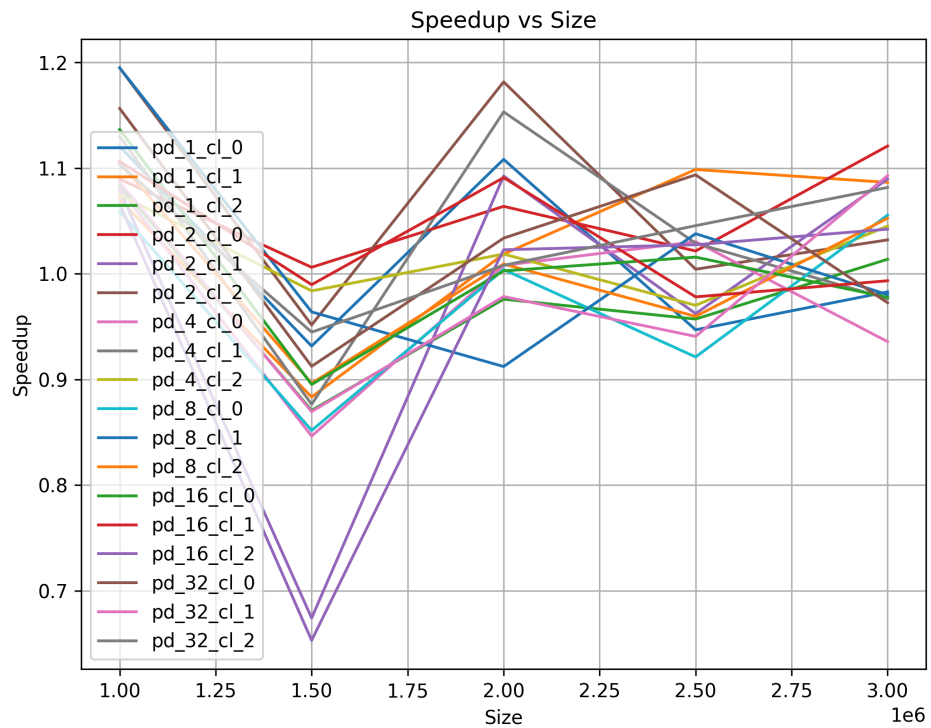
Task 2A: Software Prefetching (5 point)

- You will be **tuning** two key parameters of software prefetching:
 1. Prefetch Distance
 2. Cache Fill level
- 1. Analyze the Impact of Embedding Table Size:
 1. Increase the size of the embedding table and observe how software prefetching performance changes.

- **Execution Time vs Size:**



- **Speedup vs Size:**

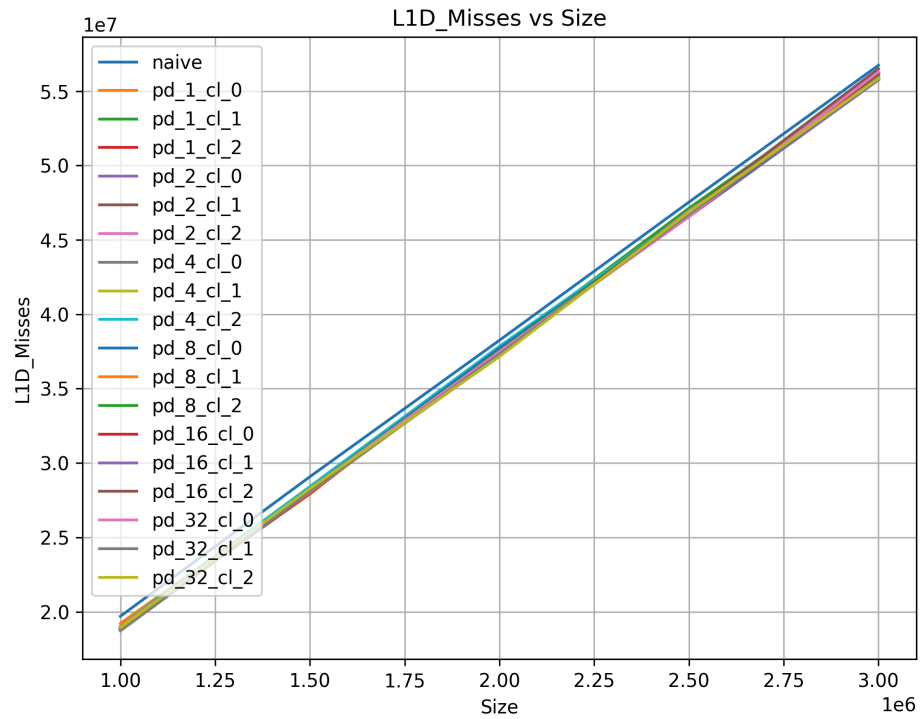


2. Optionally, you can also experiment with varying the sizes of the input array and offset array.

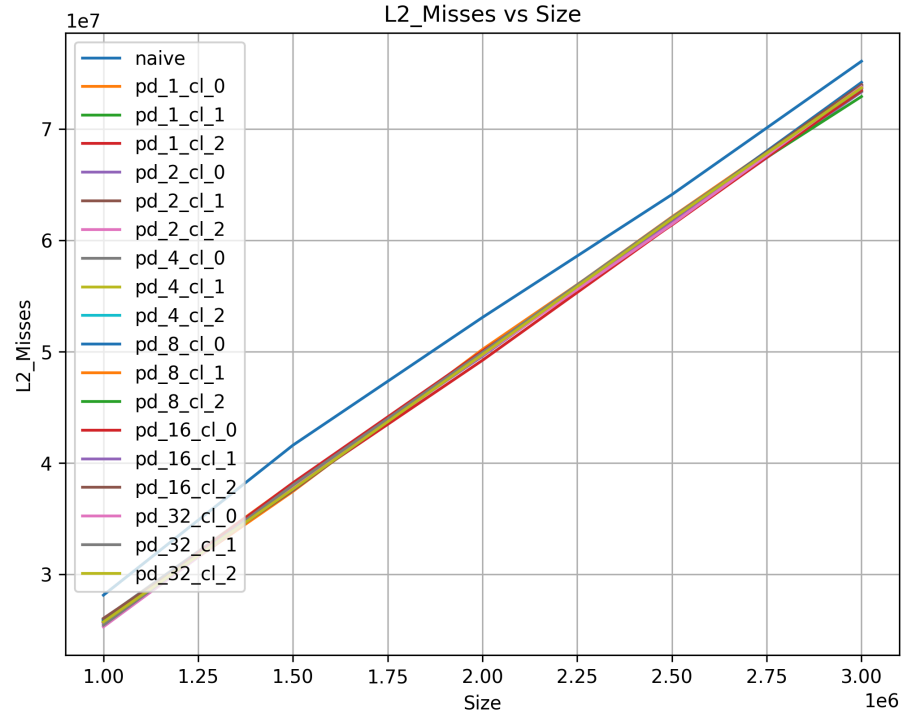
2. Analyze CPU Metrics:

1. For different embedding table sizes and different software prefetching parameters, analyze the trends in key CPU metrics (use the *perf* tool):

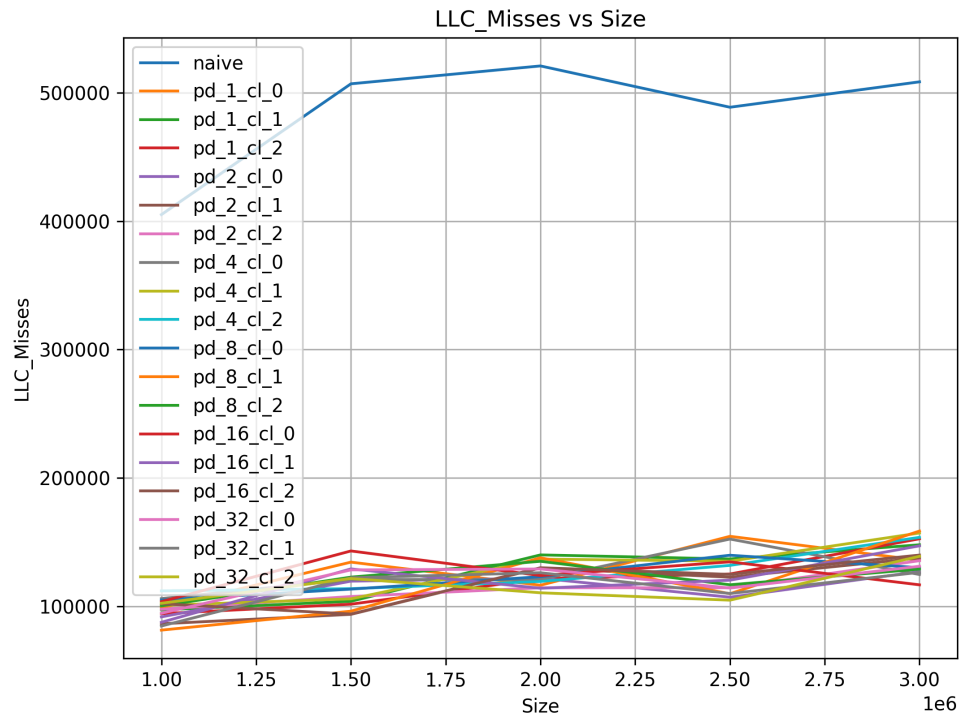
1. L1D Cache Misses:



2. L2 Cache Misses:

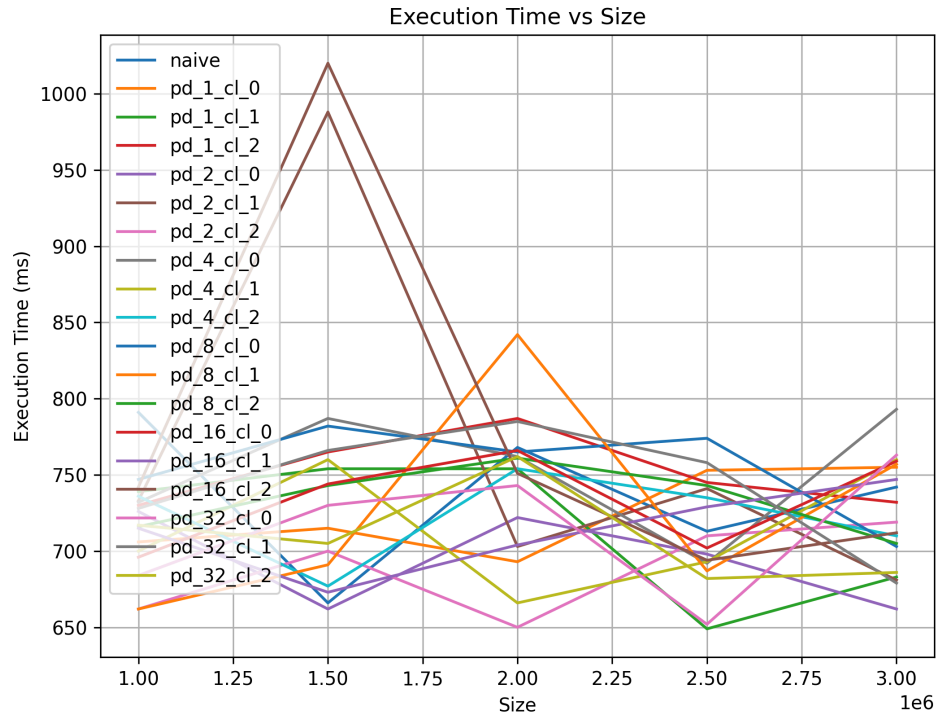


3. LLC (Last level Cache) Misses:



2. Collect Execution Time and Compute Speedup:

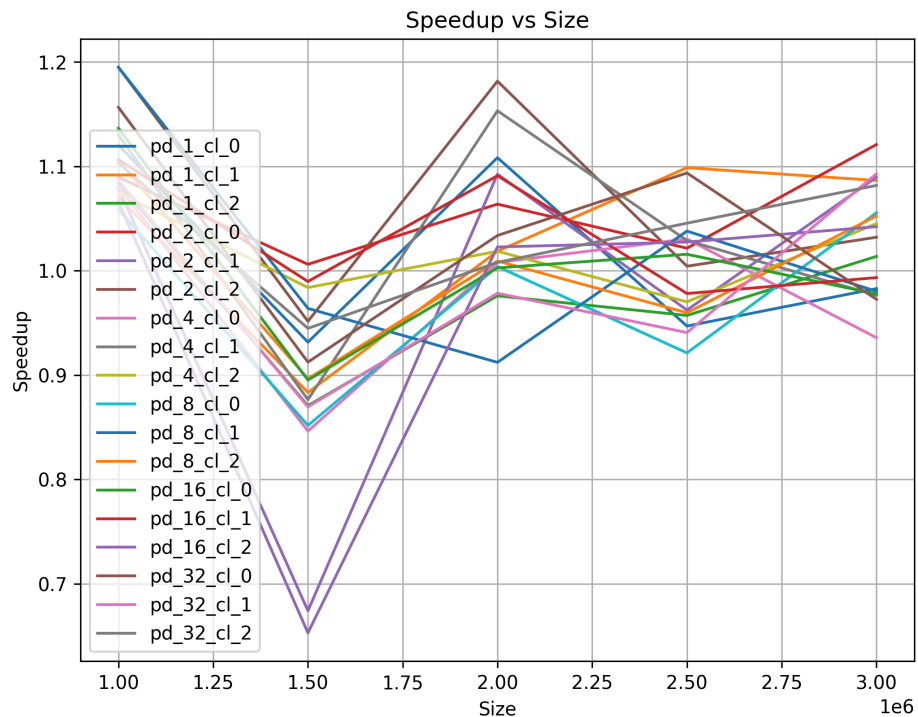
1. Measure the execution time of the embedding operation with and without software prefetching.



2. Compute the speedup as follows:

$$Speedup = \frac{ExecutionTimeWithoutSoftwarePrefetching}{ExecutionTimeWithSoftwarePrefetching}$$

3. Analyze how the speedup varies with different embedding table sizes and different software prefetching parameters.



3. Identify Optimal Parameters:

1. Determine the optimal prefetch distance and optimal cache fill level based on your observations.

- **Prefetch Distance: 16**
- **Cache fill Level: 0 (L1)**

4. Study the Effect of Hardware Prefetchers:

1. Enable and disable hardware prefetchers and analyze their impact on the embedding operation's performance.

- Didn't Try Disabling hardware prefetchers

5. Identify the values of different metrics for different embedding table sizes, prefetching distances, and cache fill levels. Analyze them carefully and state your insights for the question bellow.

1. What trend do you observe in speedup with different embedding table sizes?
 - Speedup was kind of random.
2. What is the best prefetch distance?
 - 16
3. At what cache fill level do you achieve the maximum speedup

- 0 (L1)

Task 2B: SIMD (1 point) PENDING

Task 2C: Software Prefetching + SIMD (2 point) PENDING